

Curvature and Torsion in the Continuum Theory of Lattice Defects

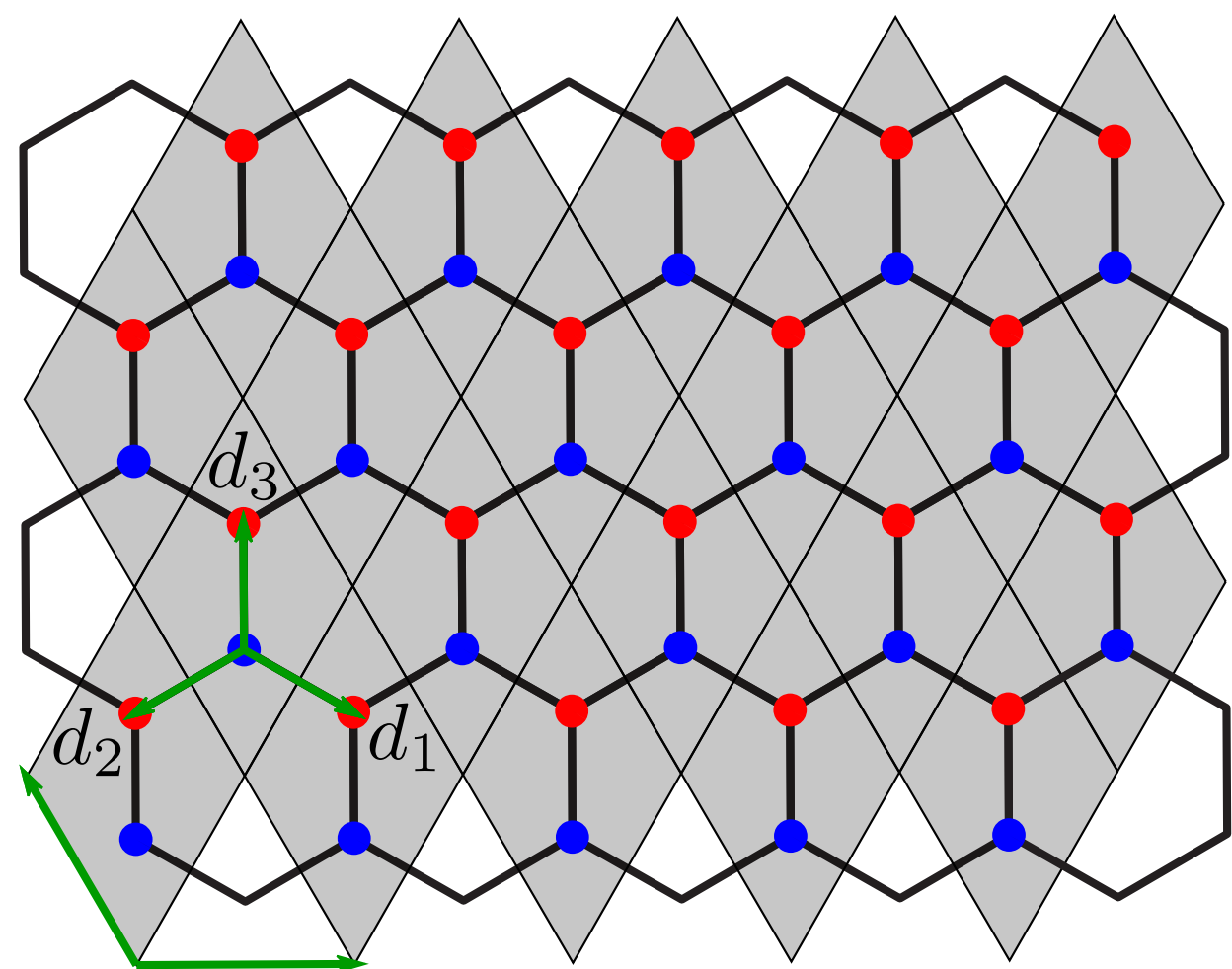
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Offen im Denken

Continuum Theory - Graphene^[1]



Tight-binding Hamiltonian

$$H = \sum_{\vec{r}} \sum_{j=1}^3 (a_{\vec{r}}^{\dagger} b_{\vec{r}+\vec{d}_j} + b_{\vec{r}+\vec{d}_j}^{\dagger} a_{\vec{r}})$$

Fourier transform

$$i\partial_t \vec{\psi}(\vec{k}) = \mathbf{h}(\vec{k}) \vec{\psi}(\vec{k})$$

$$\mathbf{h}(\vec{k}) = \begin{pmatrix} 0 & f(\vec{k}) \\ f^*(\vec{k}) & 0 \end{pmatrix}, \quad f(\vec{k}) = \sum_{j=1}^3 e^{i\vec{k} \cdot \vec{d}_j}$$

Linear dispersion around Dirac points

$$\vec{k} = \vec{K}^{(s)} + \delta\vec{k} \implies \mathbf{h}(\vec{k}) \approx -\frac{3}{2} \delta\vec{k} \cdot \vec{\sigma}$$

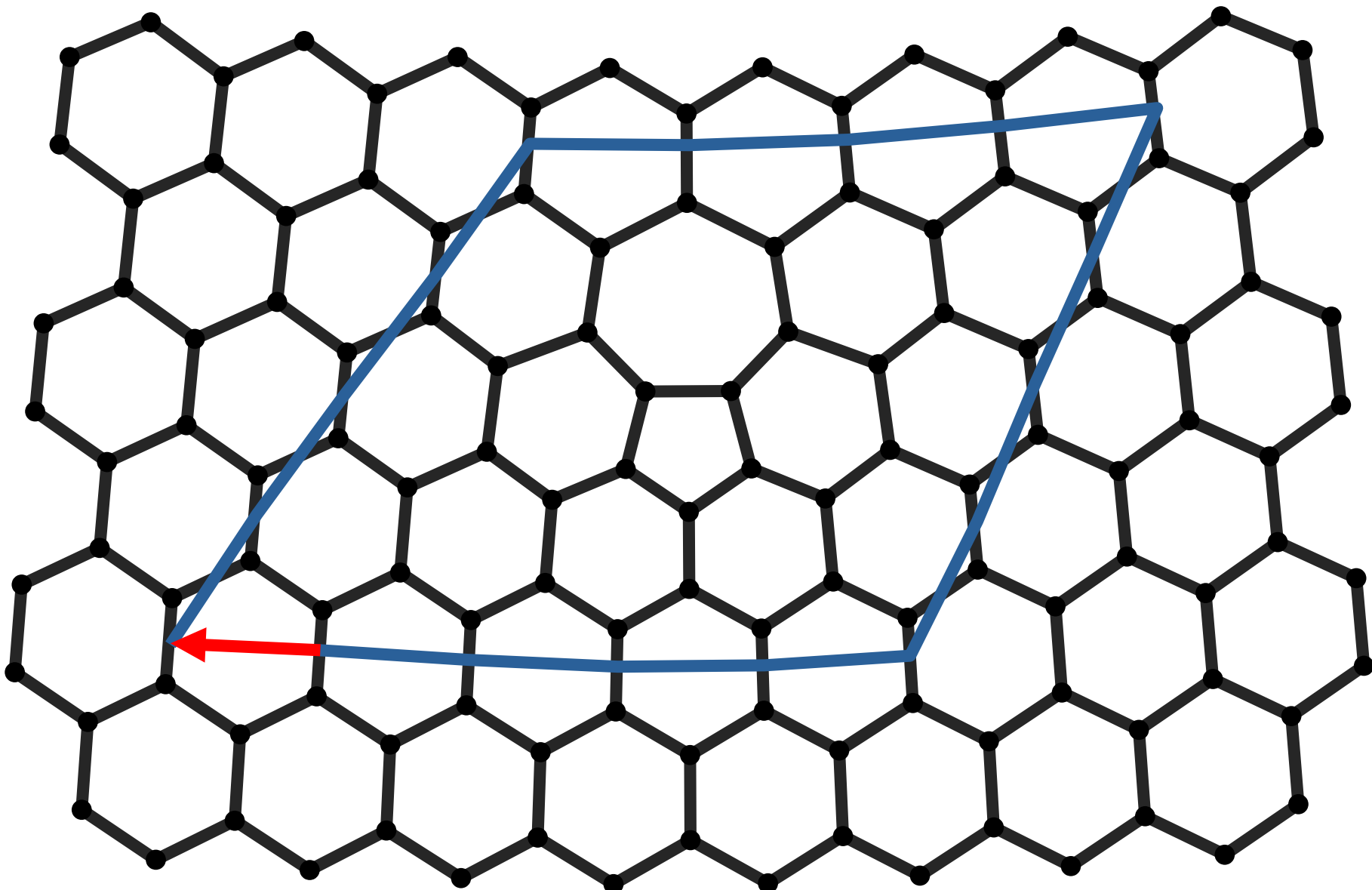
Continuum limit

$$i\partial_t \vec{\phi}(\vec{r}) = i c \sigma^i \partial_i \vec{\phi}(\vec{r}), \quad \rho = \vec{\phi}^{\dagger} \vec{\phi}, \quad \vec{j} = -c \vec{\phi}^{\dagger} \vec{\sigma} \vec{\phi}$$

$$\updownarrow$$

$$i\gamma^{\mu} \partial_{\mu} \phi = 0, \quad \gamma^{\mu} = \sigma^{\mu} \sigma^3, \quad \mu \in \{0, 1, 2\}$$

Burgers vector for a 5-7 defect^[2]



► Lattice dislocation with Burgers vector:

$$\vec{b} = -\sqrt{3} \vec{u}_x$$

Torsion^[3]

- Standard GR: torsion vanishes
- Torsion = **antisymmetric** part of the connection

$$T^{\lambda}_{\mu\nu} = \Gamma^{\lambda}_{\mu\nu} - \Gamma^{\lambda}_{\nu\mu}$$

- Decomposition of any general affine connection into symmetric part, contorsion & disformation:

$$\Gamma^{\alpha}_{\mu\nu} = \{\alpha_{\mu\nu}\} + K^{\alpha}_{\mu\nu} + L^{\alpha}_{\mu\nu}$$

- **Metric-compatibility:**

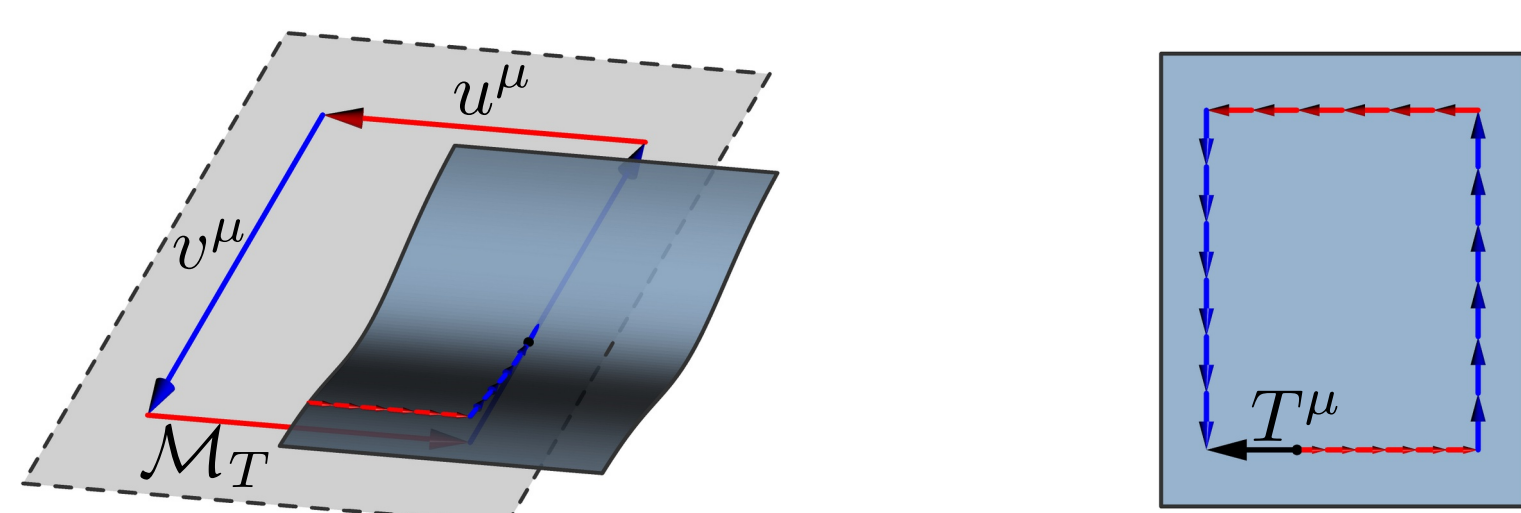
$$\nabla_{\lambda} g_{\mu\nu} = 0 \Leftrightarrow L^{\alpha}_{\mu\nu} = 0$$

- **Contorsion tensor** from torsion:

$$K^{\alpha}_{\mu\nu} = \frac{1}{2} (T^{\alpha}_{\mu\nu} - T_{\mu\nu}^{\alpha} + T_{\nu}^{\alpha}{}_{\mu})$$

- **Visual idea:** displacement vector $T^{\mu}(u, v)$ when rolling tangent plane along parallelogram:

$$T^{\mu}(u, v) = u^{\lambda} v^{\nu} T^{\mu}_{\nu\lambda}$$

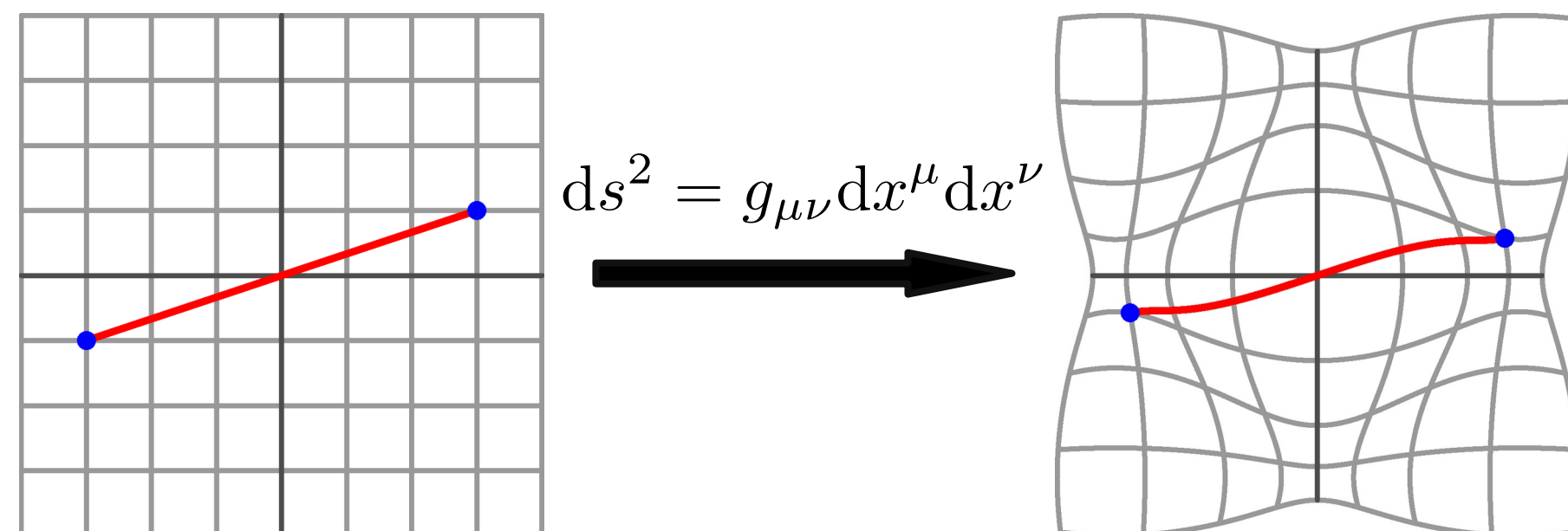


Riemannian Geometry^[3]

- Riemannian **manifold** $\mathcal{M}$ with metric $g_{\mu\nu}$ and connection $\Gamma^{\lambda}_{\mu\nu}$

Metric

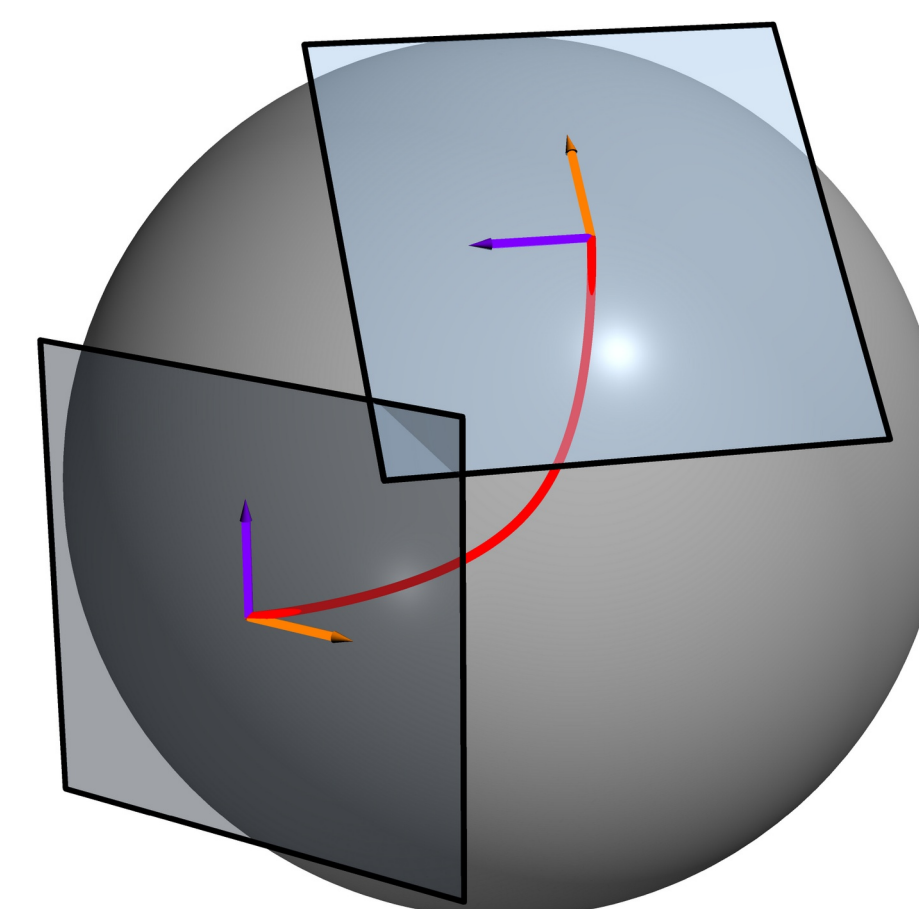
- Defines distances & angles on manifold



Connection

- Connects nearby **tangent spaces**
- Allows definition of **tensorial** derivative
- Unique torsion-free, metric-compatible connection:

Levi-Civita connection



$$\{\lambda_{\mu\nu}\} = \frac{1}{2} g^{\lambda\rho} (\partial_{\nu} g_{\rho\mu} + \partial_{\mu} g_{\rho\nu} - \partial_{\rho} g_{\mu\nu})$$

Covariant derivative

- Takes into account the **structure** of the manifold
- Transforms like a **tensor**

$$\nabla_{\mu} v^{\nu} = \partial_{\mu} v^{\nu} + \Gamma^{\nu}_{\mu\lambda} v^{\lambda}$$

Autoparallels

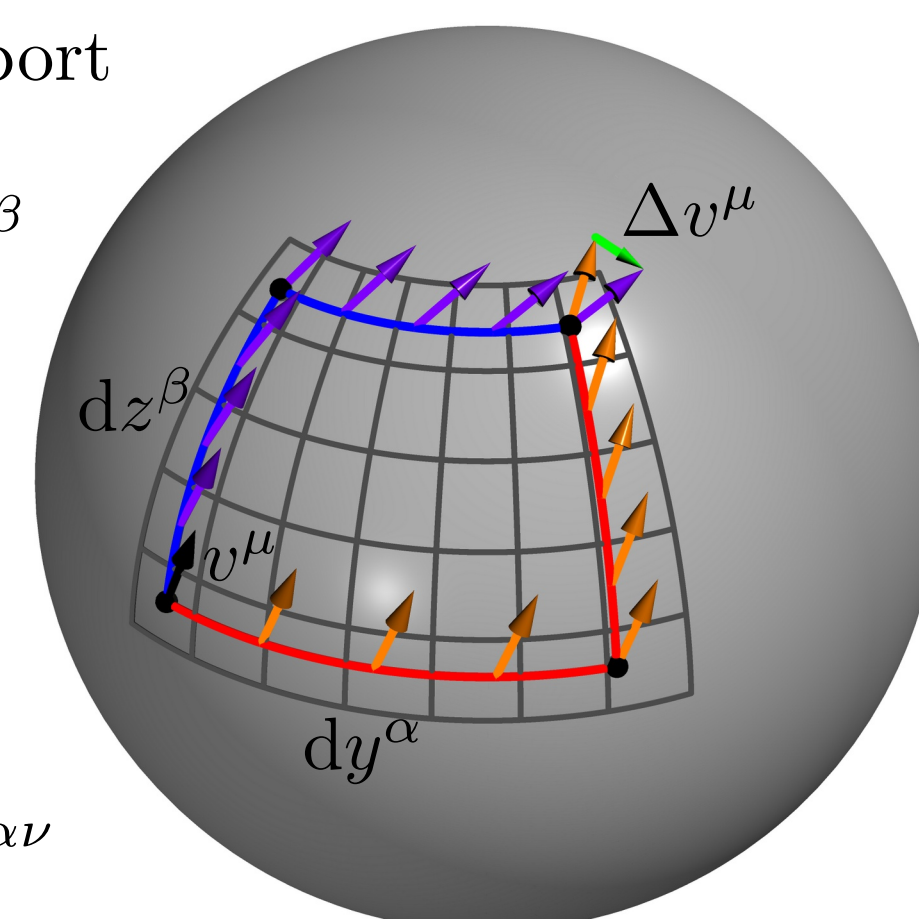
- Tangent vector is parallel-transported

$$u^{\mu} \nabla_{\mu} u^{\alpha} = 0$$

Curvature

- how does parallel transport depend on the path?
- $\Delta v^{\mu} = R_{\alpha\beta}{}^{\mu}{}_{\nu} v^{\nu} dy^{\alpha} dz^{\beta}$
- Fully determined by the connection:

$$R_{\alpha\beta}{}^{\mu}{}_{\nu} = +\partial_{\alpha} \Gamma^{\mu}_{\beta\nu} - \partial_{\beta} \Gamma^{\mu}_{\alpha\nu} + \Gamma^{\mu}_{\alpha\lambda} \Gamma^{\lambda}_{\beta\nu} - \Gamma^{\mu}_{\beta\lambda} \Gamma^{\lambda}_{\alpha\nu}$$



Metric $\leftrightarrow$ torsion mapping in 2D

- Physical mapping between:
 - **Classical** Manifold $\mathcal{M}_g(g_{ij}, \{k_{ij}^k\})$
 - **Purely torsionful** Manifold $\mathcal{M}_T(\delta_{ij}, K^k_{ij})$
- Coordinate choice: $g_{\mu\nu} = \delta_{\mu\nu} e^{2\phi(x,y)}$.
- Antisymmetry: $T^k_{ij} = b^k \varepsilon_{ij}$
- Demand equal **curvature** of both manifolds:

$$R[\mathcal{M}_g] = R[\mathcal{M}_T],$$

- leads to solution for torsion vector b^i as function of metric ϕ and arbitrary scalar field χ :

$$b^i = \varepsilon_{ik} \partial_k \phi + \partial_i \chi$$

- for $\chi = 0$: **autoparallels** of both manifolds equivalent up to reparameterization
- χ corresponds to local **rotations** of the frame field

Conclusion

- Autoparallels and curvature can be mapped
- Different distance measure
- Other tangent spaces and tensors

References

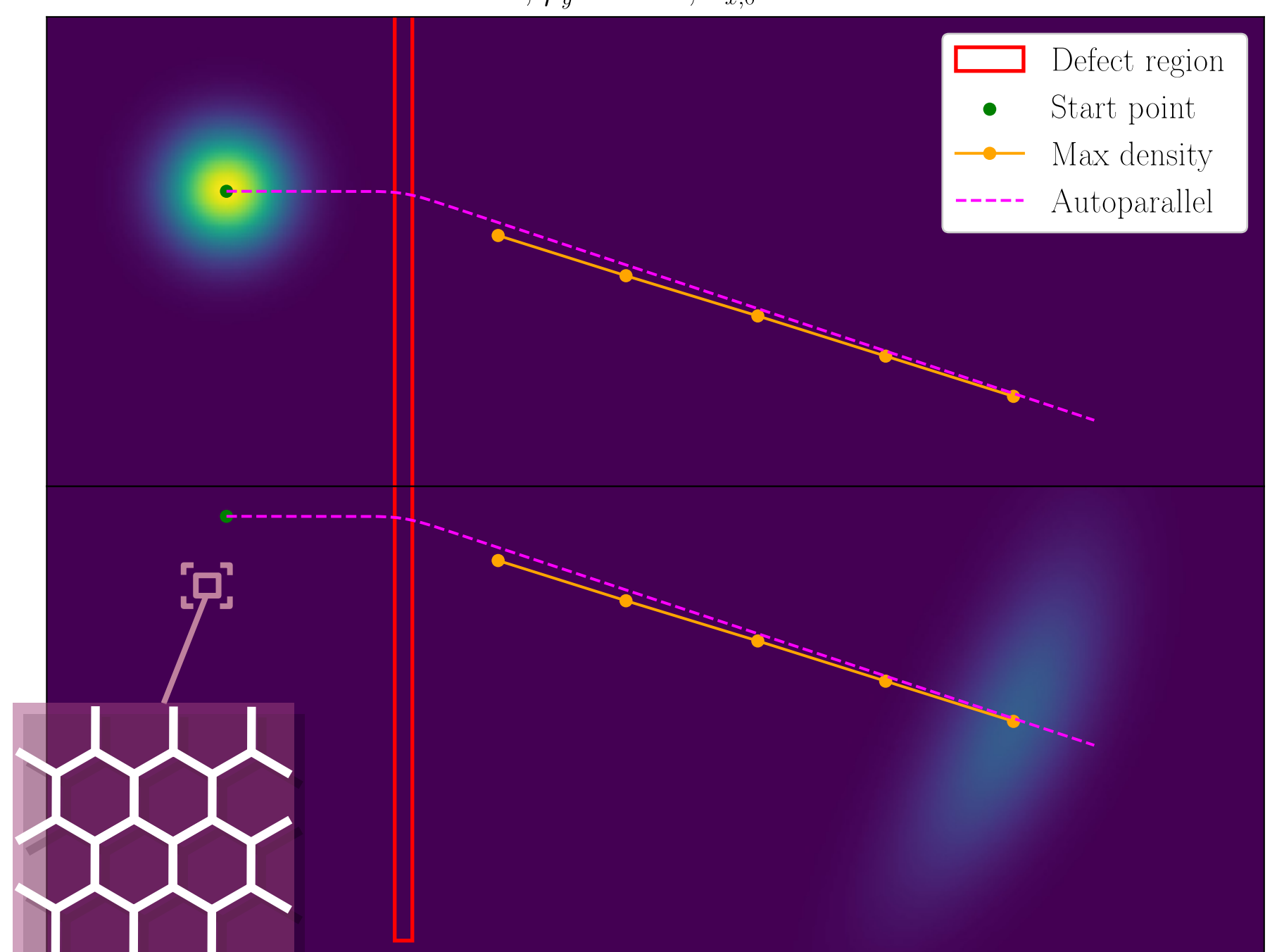
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- [2] Bhatt MD, Kim H, Kim G. Various defects in graphene: a review. *RSC Adv.* **12**, 21520 (2022).
- [3] Heisenberg L. A systematic approach to generalisations of General Relativity. *Phys. Rep.* **796**, 1 (2019).
- [4] de Juan F, Cortijo A, Vozmediano MAH. Dislocations and torsion in graphene. *Nucl. Phys. B* **828**, 625 (2010).

Tight-binding numerics

- **vector representation** of wavefunction at discrete lattice sites
- Hamiltonian is (sparse) **Matrix** acting on this vector to generate time evolution:

$$i\partial_t \vec{\psi} = H \vec{\psi} \implies \vec{\psi}(t) = e^{-iHt} \vec{\psi}_0$$

$$C = 1, \rho_y = 0.385, k_{x,0} = 0.121$$



Torsion model

- $\mathcal{M}_T$ with $b^1(x)$ and $b^2 \equiv 0$
- Autoparallel equation:

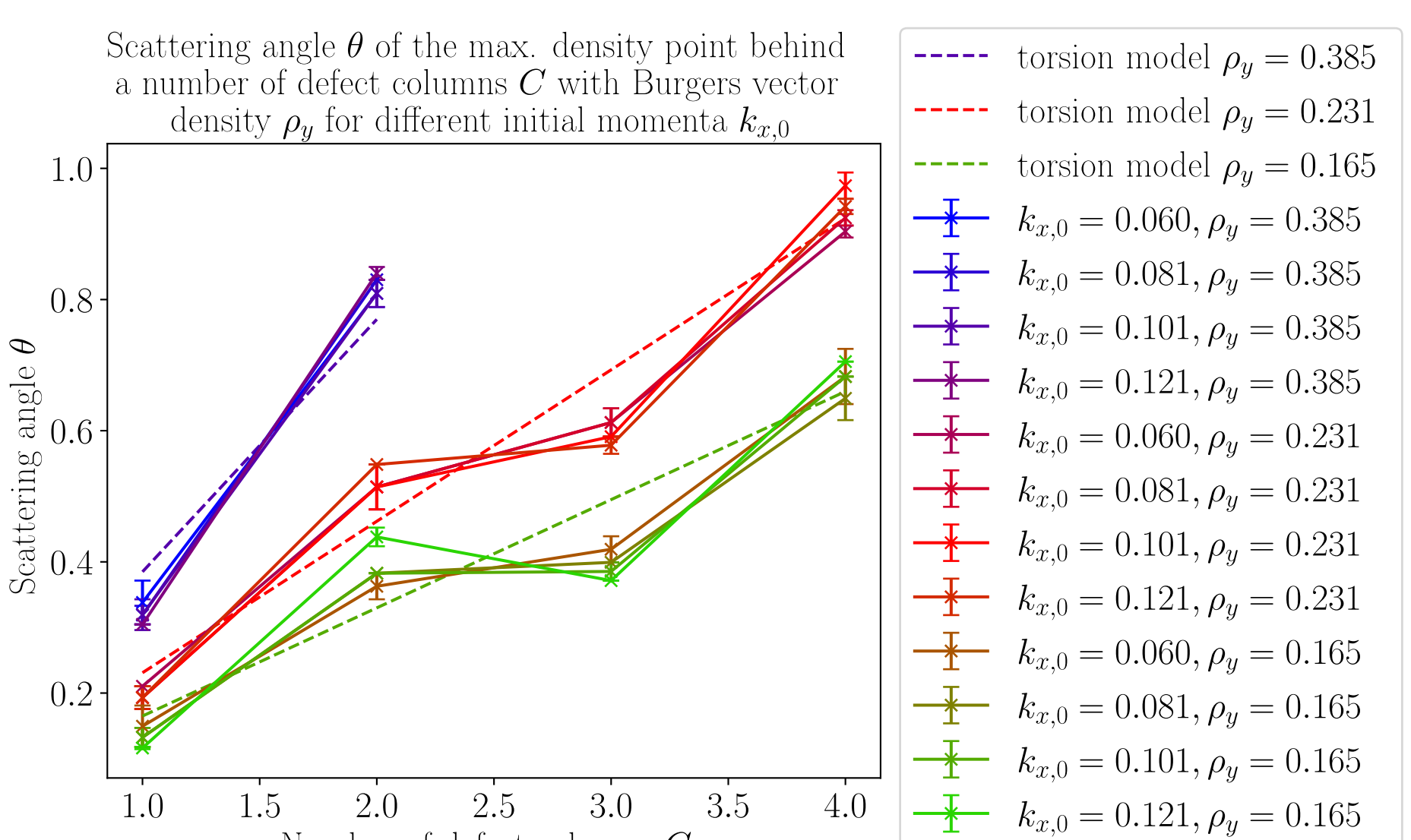
$$\ddot{x} = -\dot{x}\dot{y}b^1, \quad \ddot{y} = \dot{x}^2 b^1$$

- Allows calculating the **scattering angle**:

$$\theta = \int b^1(x) dx$$

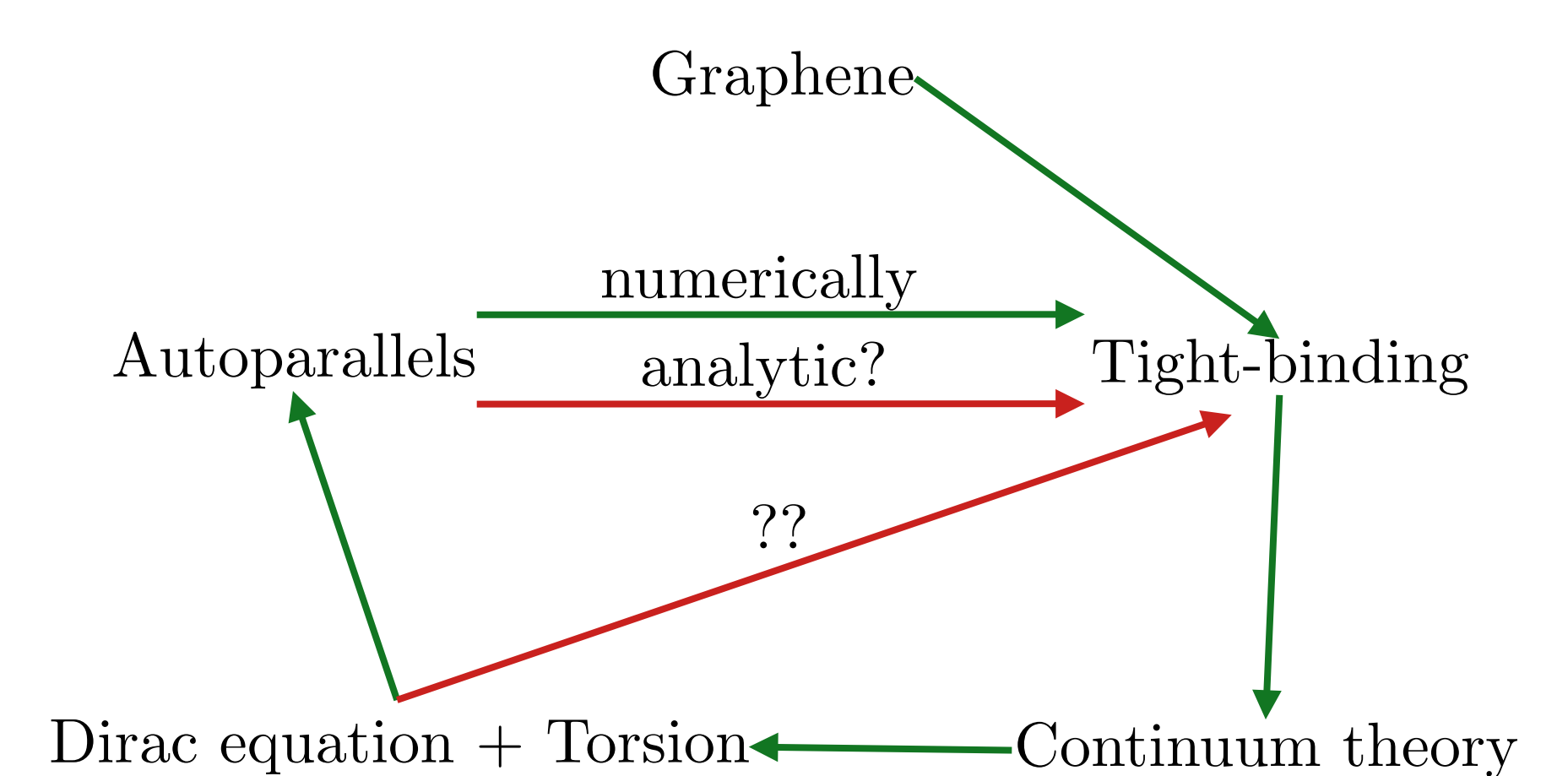
- Interpreting torsion as Burgers vector density:

$$\rho_y C = \frac{C \cdot (\vec{b})_x}{\Delta y} \approx \frac{\iint b^1 dx dy}{\Delta y} = \int b^1 dx = \theta$$



- Torsion model predicts linear relation $\theta = \rho_y C$
- Tight-binding simulations match prediction from torsion model with autoparallels

Conclusion & Outlook^[4]



- Can the torsion interpretation of defects via autoparallels be shown analytically?
- Is it possible to describe the tight-binding results with the Dirac equation coupled to torsion?
- Does the physical mapping between torsion and metric apply to the description via torsion?